

helixcluster

You are an expert SVG author. Output ONLY one valid self-contained <svg> (SVG 1.1, xmlns set). No HTML, no <script>, no external refs, no markdown, no prose. SVG source only. Reproduce every element EXACTLY at the coordinates given. Do not re-layout, move, or resize anything. Draw CONNECTORS FIRST, then boxes, then text (so boxes sit above lines).

Root: <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 960 680" font-family="'Inter',system-ui,sans-serif">.

First child MUST be this <style> (verbatim):

```
<style>
svg{--ink:#1e293b;--muted:#475569;--line:#94a3b8;--
accent:#a31e39;--panel:#f8fafc;--panel2:#eef2f7;--tint:#f4ddel;--
good:#dcfce7;--goodln:#16a34a;--goodink:#14532d;}
@media(prefers-color-scheme:dark){svg{--ink:#e2e8f0;--
muted:#cbd5e1;--line:#64748b;--accent:#ff6b81;--panel:#1e293b;--
panel2:#273449;--tint:#3a2530;--good:#14321f;--goodln:#4ade80;--
goodink:#bbf7d0;}}
.box{fill:var(--panel);stroke:var(--line);stroke-width:1.5;}
.accent{fill:var(--panel);stroke:var(--accent);stroke-width:2;}
.tint{fill:var(--tint);stroke:var(--accent);stroke-width:1.5;}
.good{fill:var(--good);stroke:var(--goodln);stroke-width:1.5;}
.dash{fill:var(--panel2);stroke:var(--line);stroke-
width:1.5;stroke-dasharray:5 4;}
.t{fill:var(--ink);font-size:15px;font-weight:700;text-
anchor:middle;}
.s{fill:var(--muted);font-size:11px;text-anchor:middle;}
.gt{fill:var(--goodink);font-size:15px;font-weight:700;text-
anchor:middle;}
.h{fill:var(--ink);font-size:20px;font-weight:700;}
.lbl{fill:var(--muted);font-size:12px;font-weight:600;}
.note{fill:var(--muted);font-size:11px;text-anchor:middle;}
.conn{stroke:var(--line);stroke-width:1.75;fill:none;}
.strike{stroke:var(--accent);stroke-width:2.5;}
</style>
```

Then <defs><marker id="a" markerWidth="9" markerHeight="9" refX="7" refY="3" orient="auto"><path d="M0,0 L7,3 L0,6 Z" fill="var(--line)"/></marker></defs>.

How to draw each BOX row “(x, y, w, h, class, title, sub)”: <rect class="{class}" x y width height rx="10"/>, then <text class="{t or gt}" x="{x+w/2}" y="{y+34}">{title}</text>; if sub is non-empty add <text class="s" x="{x+w/2}" y="{y+54}">{sub}</text>; if sub is empty put title at y+{h/2+5}. Use class gt for the title only when the box class is good, otherwise class t. Each CONNECTOR line: <path class="conn" marker-end="url(#a)" d="{d}"/> using the given d verbatim. Each ANNOTATION row “(x, y, class, text)”: <text class="{class}" x="{x}" y="{y}">{text}</text>. Class lbl renders left-aligned; class note renders centered — use the class exactly as given. Each STRIKE row “(x1, y1, x2, y2)”: <line class="strike" x1 y1 x2 y2/> (used to cross out a box). Omit the STRIKE section if the spec says none. If a section says “none”, omit it entirely.

DIAGRAM SPEC: TITLE: <text class="h" x="32" y="44">HelixCluster – seven-layer stack</text>

ANNOTATIONS (x, y, class, text): - 40, 88, lbl, “14 control-plane” - 40, 106, lbl, “microservices” - 40, 548, lbl, “heterogeneous” - 40, 566, lbl, “nodes T1-T8” - 210, 632, note, “Node tiers T1 (datacenter GPU) → T8 (handheld), unified under one control plane”

BOXES (x, y, w, h, class, title, sub): - 210, 78, 560, 54, box, “L7 · Federation & Observability”, “” - 210, 140, 560, 54, box, “L6 · Security & Attestation (SPIFFE, PQ-E2EE)”, “” - 210, 202, 560, 54, box, “L5 · Sessions & Interactive Terminal”, “” - 210, 264, 560, 54, box, “L4 · AI Inference Routing”, “” - 210, 326, 560, 54, accent, “L3 · Omega Scheduler (two-level)”, “” - 210, 388, 560, 54, box, “L2 · Consensus & Membership (Raft + SWIM)”, “” - 210, 450, 560, 54, box, “L1 · Node Runtime & Transport (gRPC, WireGuard)”, “” - 210, 512, 560, 54, box, “L0 · Hardware Substrate (GPU → edge SBC)”, “”

CONNECTORS: none